USER INFORMATION
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LEVOFLOXOL®
LEVOFLOXACIN 5 mg/mL INTRAVENOUS INFUSION

Composition:
Each 100 mL solution contains :-
Levofloxacin 500 mg as active ingredient.

Product Description:
LEVOFLOXOL® is a sterile and non-pyrogenic bactericidal quinolone antibiotic containing Levofloxacin 5 mg/mL. The solution is a clear greenish-yellow solution.

Pharmacodynamic:
Levofloxacin is a synthetic antibacterial agent of the fluoroquinolone class and is the S (-) enantiomer of the racemic drug substance Ofloxacin.
Mechanism of action
As a fluoroquinolone antibacterial agent, Levofloxacin acts on the DNA-DNA-gyrase complex and topoisomerase IV.
PK/PD relationship
The degree of the bactericidal activity of Levofloxacin depends on the ratio of the maximum concentration in serum (Cmax) or the area under the curve (AUC) and the minimal inhibitory concentration (MIC).
Mechanism of resistance
Resistance to Levofloxacin is acquired through a stepwise process by target site mutations in both type II topoisomerases, DNA gyrase and topoisomerase IV. Other resistance mechanisms such as permeation barriers (common in *Pseudomonas aeruginosa*) and efflux mechanisms may also affect susceptibility to Levofloxacin.
Cross-resistance between Levofloxacin and other fluoroquinolones is observed. Due to the mechanism of action, there is generally no cross-resistance between Levofloxacin and other classes of antibacterial agents.
Breakpoints
The European Committee on Antimicrobial Susceptibility Testing (EUCAST) recommended MIC breakpoints for Levofloxacin, separating susceptible from susceptible increased exposure organisms and susceptible increased exposure from resistant organisms are presented in the below table for MIC testing (mg/L).

EUCAST clinical MIC breakpoints for Levofloxacin (Version 2.0; 2021-01-01):

Pathogen	Susceptible	Resistant
Enterobacterales	≤0.5 mg/L	>1 mg/L
Pseudomonas spp.	≤0.001 mg/L	>1 mg/L
Acinetobacter spp.	≤0.5 mg/L	>1 mg/L
Staphylococcus spp.	≤0.0.001 mg/L	>1 mg/L
Enterococcus spp.[1]	≤4 mg/L	>4 mg/L
Streptococcus groups A, B, C and G	≤0.001 mg/L	>2 mg/L
Streptococcus pneumoniae	≤0.001 mg/L	>2 mg/L
Haemophilus influenzae	≤0.06 mg/L	>0.06 mg/L
Moraxella catarrhalis	≤0.125 mg/L	>0.125 mg/L
Helicobacterpylori	≤1 mg/L	>1 mg/L
P. multocida	≤0.06 mg/L	>0.06 mg/L
Aerococcussanguinicola and *urinae*[1]	≤2 mg/L	>2 mg/L
K. kingae	≤0.125 mg/L	>0.125 mg/L
Aeromonas spp.	≤0.5 mg/L	>1 mg/L
Bacillus spp. (Not anthracis)	≤0.0.001 mg/L	>1 mg/L
PK-PD breakpoints (non-species related)	≤0.5 mg/L	>1 mg/L

[1] Uncomplicated urinary tract infections (UTI) only

The prevalence of resistance may vary geographically and with time for selected species and local information on resistance is desirable, particularly when treating severe infections. As necessary, expert advice should be sought when the local prevalence of resistance is such that the utility of the agent in at least some types of infections is questionable.

Commonly susceptible species
Aerobic Gram-positive bacteria
Bacillus anthracis
Staphylococcus aureusmethicillin-susceptible
Staphylococcus saprophyticus
Streptococci, group C and G
Streptococcus agalactiae
Streptococcus pneumoniae
Streptococcus pyogenes

Aerobic Gram-negative bacteria
Eikenella corrodens
Haemophilus influenzae
Haemophilus para-influenzae
Klebsiella oxytoca
Moraxella catarrhalis
Pasteurella multocida
Proteusvulgaris
Providencia rettgeri

Anaerobic bacteria
Peptostreptococcus

Other
Chlamydophila pneumoniae
Chlamydophila psittaci
Chlamydia trachomatis
Legionella pneumophila
Mycoplasma pneumoniae
Mycoplasma hominis
Ureaplasma urealyticum

Species for which acquired resistance may be a problem
Aerobic Gram-positive bacteria
Enterococcus faecalis
Staphylococcus aureus methicillin-resistant[#]
Coagulase negative Staphylococcus spp

Aerobic Gram-negative bacteria
Acinetobacter baumannii
Enterobacter aerogenes
Escherichia coli
Morganella morganii
Providencia stuartii
Serratia marcescens
Citrobacter freundii
Enterobactercloacae
Klebsiella pneumoniae
Proteus mirabilis
Pseudomonas aeruginosa

Anaerobic bacteria
Bacteroides fragilis

Inherently Resistant Strains
Aerobic Gram-positive bacteria
Enterococcus faecium

[#] *Methicillin-resistant S. aureus is very likely to possess co-resistance to fluoroquinolones, including Levofloxacin.*

Pharmacokinetics:
Absorption
Orally administered Levofloxacin is rapidly and almost completely absorbed with peak plasma concentrations being obtained within 1 - 2 hours. The absolute bioavailability is 99 - 100 %. Food has little effect on the absorption of Levofloxacin. Steady state conditions are reached within 48 hours following a 500 mg once or twice daily dosage regimen.
Distribution
Approximately 30 - 40 % of Levofloxacin is bound to serum protein. The mean volume of distribution of Levofloxacin is approximately 100 L after single and repeated 500 mg doses, indicating widespread distribution into body tissues.
Penetration into tissues and body fluids:
Levofloxacin has been shown to penetrate into bronchial mucosa, epithelial lining fluid, alveolar macrophages, lung tissue, skin (blister fluid), prostatic tissue and urine. However, Levofloxacin has poor penetration into cerebro-spinal fluid.
Biotransformation
Levofloxacin is metabolised to a very small extent, the metabolites being desmethyl-Levofloxacin and Levofloxacin N-oxide. These metabolites account for < 5 % of the dose excreted in urine. Levofloxacin is stereo chemically stable and does not undergo chiral inversion.
Elimination
Following oral and intravenous administration of Levofloxacin, it is eliminated relatively slowly from the plasma (t½: 6 – 8 hours). Excretion is primarily by the renal route (> 85 % of the administered dose). The mean apparent total body clearance of Levofloxacin following a 500 mg single dose was 175 +/-29.2 mL/min. There are no major differences in the pharmacokinetics of Levofloxacin following intravenous and oral administration, suggesting that the oral and intravenous routes are interchangeable.
Linearity
Levofloxacin obeys linear pharmacokinetics over a range of 50 to 1000 mg.
Special populations
Subjects with renal insufficiency
The pharmacokinetics of Levofloxacin are affected by renal impairment. With decreasing renal function renal elimination and clearance are decreased, and elimination half-lives increased as shown in the table below:
Pharmacokinetics in renal insufficiency following single oral 500 mg dose

Clcr [mL/min]	< 20	20 - 49	50 - 80
ClR [mL/min]	13	26	57
t1/2 [h]	35	27	9

Elderly subjects
There are no significant differences in Levofloxacin pharmacokinetics between young and elderly subjects, except those associated with differences in creatinine clearance.
Gender differences
Separate analysis for male and female subjects showed small to marginal gender differences in Levofloxacin pharmacokinetics. There is no evidence that these gender differences are of clinical relevance.

Indications:
Treatment of adults (more than or equal to 18 years) with mild, moderate and severe infections caused by susceptible strains of the designated microorganisms in the conditions listed as follows. LEVOFLOXOL® is indicated when I.V. administration offers a route of administration advantageous to the patient (e.g., patient cannot tolerate an oral dosage form).
- Acute maxillary sinusitis due to *Streptococcus pnuemoniae, Haemophilus influenzae* or *Moraxella catarrhalis*.
- *Acute bacterial exacerbation of chronic bronchitis due to *Staphylococcus aureus, Streptococcus pnuemoniae, Haemophilus influenzae, Haemophilus parainfluenzae* or *Moraxella catarrhalis*.
- *Community-acquired pneumonia due to *Staphylococcus aureus, Streptococcus pnuemoniae, Legionella pnuemophila* or *Mycoplasma pneumoniae*.
- Uncomplicated skin and skin structure infections (mild to moderate) including abscesses, cellulitis, furuncles, impetigo, pyoderma, wound infections, due to *Staphylococcus aureus* or *Streptococcus pyogenes*.
- *Complicated urinary tract infections (mild to moderate) due to *Enterococcus faecalis, Enterobacter cloacae, Escherichia coli, Klebsiella pnuemoniae, Proteus mirabilis* or *Pseudomonas aeruginosa*.
- Acute pyelonephritis (mild to moderate) caused by *Escherichia coli*.

Appropriate culture and susceptibility tests should be performed before treatment in order to isolate and identify organisms causing the infection and to determine their susceptibility to Levofloxacin. Therapy with Levofloxacin may be initiated before results of these tests are known; once results become available, appropriate therapy should be selected.
As with other drugs in this class, some strains of *Pseudomonas aeruginosa* may develop resistance fairly during treatment with Levofloxacin. Culture and susceptibility testing performed periodically during therapy will provide information about the continued susceptibility of the pathogens to the antimicrobial agent and also the possible emergence of bacterial resistance.

* LEVOFLOXOL® should be only used:
When *Pseudomonas* is considered AND patient is allergic to antipseudomonal penicillins/cephalosporins;
For resistant organisms with no other alternative antibiotics available.

Recommended Dosage:
LEVOFLOXOL® is administered by slow intravenous infusion once or twice daily. The dosage depends on the type and severity of the infection and the susceptibility of the presumed causative pathogen. Treatment with Levofloxacin after initial use of the intravenous preparation may be completed with an appropriate oral presentation according to the SPC for the film-coated tablets and as considered appropriate for the individual patient. Given the bioequivalence of the parenteral and oral forms, the same dosage can be used.

Posology
The following dose recommendations can be given for Levofloxacin
Dosage in patients with normal renal function (creatinine clearance > 50 mL/min)

Indication	Daily dose regimen (according to severity)	Total duration of treatment[1] (according to severity)
Community-acquired pneumonia	500 mg once or twice daily	7 - 14 days
Acute pyelonephritis	500 mg once daily	7 - 10 days
Complicated urinary tract infections	500 mg once daily	7 - 14 days
Chronic bacterial prostatitis	500 mg once daily	28 days
Complicated skin and soft tissue infections	500 mg once or twice daily	7 - 14 days
Inhalation anthrax	500 mg once daily	8 weeks

[1] Treatment duration includes intravenous plus oral treatment. The time to switch from intravenous to oral treatment depends on the clinical situation but is normally 2 to 4 days.

Special populations
Impaired renal function (creatinine clearance ≤ 50 mL/min)

Dose regimen	250 mg/24 h	500 mg/24 h	500 mg/12 h
Creatinine clearance	first dose: 250 mg	first dose: 500 mg	first dose: 500 mg
50 - 20 mL/min	then: 125 mg/24 h	then: 250 mg/24 h	then: 250 mg/12 h
19 - 10 mL/min	then: 125 mg/48 h	then: 125 mg/24 h	then: 125 mg/12 h
< 10 mL/min (including haemodialysis and CAPD)[1]	then: 125 mg/48 h	then: 125 mg/24 h	then: 125 mg/12 h

[1] No additional doses are required after haemodialysis or continuous ambulatory peritoneal dialysis (CAPD).

Impaired liver function
No adjustment of dose is required since Levofloxacin is not metabolised to any relevant extent by the liver and is mainly excreted by the kidneys.
Elderly population
No adjustment of dosage is required in the elderly, other than that imposed by consideration of renal function.

Paediatric population
Levofloxacin is contraindicated in children and growing adolescents.
Method of administration
LEVOFLOXOL® is only intended for slow intravenous infusion; it is administered once or twice daily. The infusion time must be at least 30 minutes for 250 mg or 60 minutes for 500 mg Levofloxacin solution for infusion.

Route of Administration : Intravenous (IV)

Contraindications:
LEVOFLOXOL® must not be used:
1. Patients hypersensitivity to Levofloxacin or any other quinolone.
2. Patients with history of tendon disorders related of fluoroquinolone administrations.
3. Patients with epilepsy.
4. Children or growing adolescents.
5. Breast-feeding women.
6. During pregnancy.

Warning and Precautions
The use of Levofloxacin should be avoided in patients who have experienced serious adverse reactions in the past when using quinolone or fluoroquinolone containing products. Treatment of these patients with Levofloxacin should only be initiated in the absence of alternative treatment options and after careful benefit/risk assessment.
Risks of resistance
Methicilin-resistant *S. aureus* are very likely to possess co-resistance to fluoroquinolones, including Levofloxacin. Therefore, Levofloxacin is not recommended for the treatment of known or suspected MRSA infections unless laboratory results have confirmed susceptibility of the organism to Levofloxacin (and commonly recommended antibacterial agents for the treatment of MRSA-infections are considered inappropriate).
Resistance to fluoroquinolones of *E. coli* – the most common pathogen involved in urinary tract infections – varies across the European Union. Prescribers are advised to take into account the local prevalence of resistance in *E. coli* to fluoroquinolones.
Inhalation Anthrax
Use in humans is based on in vitro *Bacillus anthracis* susceptibility data and on animal experimental data together with limited human data. Treating physicians should refer to national and/or international consensus documents regarding the treatment of anthrax.
Prolonged, disabling and potentially irreversible serious adverse drug reactions
Very rare cases of prolonged (continuing months or years), disabling and potentially irreversible serious adverse drug reactions affecting different, sometimes multiple body systems (musculoskeletal, nervous, psychiatric and senses) have been reported in patients receiving fluoroquinolones irrespective of their age and pre-existing risk factors. Levofloxacin should be discontinued immediately at the first signs or symptoms of any serious adverse reaction and patients should be advised to contact their prescriber for advice.
Infusion Time
The recommended infusion time of at least 30 minutes for 250 mg or 60 minutes for 500 mg LEVOFLOXOL® should be observed. It is known for Ofloxacin that during infusion tachycardia and a temporary decrease in blood pressure may develop. In rare cases, as a consequence of a profound drop in blood pressure, circulatory collapse may occur. Should a conspicuous drop in blood pressure occur during infusion of Levofloxacin, (l-isomer of Ofloxacin) the infusion must be halted immediately.
Tendonitis and tendon rupture
Tendinitis and tendon rupture (especially but not limited to Achilles tendon), sometimes bilateral, may occur as early as within 48 hours of starting treatment with fluoroquinolones and have been reported to occur even up to several months after discontinuation of treatment. The risk of tendinitis and tendon rupture is increased in older patients (above 60 years of age), with renal impairment, patients with solid organ transplants, and those treated concurrently with corticosteroids in patients receiving daily doses of 1000 mg Levofloxacin. Therefore, concomitant use of corticosteroids should be avoided.
At the first sign of tendinitis (e.g. painful swelling, inflammation) the treatment with Levofloxacin should be discontinued and alternative treatment should be considered. The affected limb(s) should be appropriately treated (e.g. immobilisation). Corticosteroids should not be used if signs of tendinopathy occur.
Clostridium difficile-associated disease
Diarrhoea, particularly if severe, persistent and/or bloody, during or after treatment with Levofloxacin (including several weeks after treatment), may be symptomatic of Clostridium difficile-associated disease (CDAD). CDAD may range in severity from mild to life threatening, the most severe form of which is *pseudomembranous colitis*. It is therefore important to consider this diagnosis in patients who develop serious diarrhoea during or after treatment with Levofloxacin. If CDAD is suspected or confirmed, Levofloxacin should be stopped immediately and appropriate treatment initiated without delay. Anti-peristaltic medicinal products are contraindicated in this clinical situation.
Patients predisposed to seizures
Quinolones may lower the seizure threshold and may trigger seizures. Levofloxacin is contraindicated in patients with a history of epilepsy and, as with other quinolones, should be used with extreme caution in patients predisposed to seizures or concomitant treatment with active substances that lower the cerebral seizure threshold, such as theophylline. In case of convulsive seizures, treatment with Levofloxacin should be discontinued.
Patients with G-6-phosphate dehydrogenase deficiency
Patients with latent or actual defects in glucose-6-phosphate dehydrogenase activity may be prone to haemolytic reactions when treated with quinolone antibacterial agents. Therefore, if Levofloxacin has to be used in these patients, potential occurrence of haemolysis should be monitored.
Patients with renal impairment
Since Levofloxacin is excreted mainly by the kidneys, the dose of Levofloxacin solution for infusion should be adjusted inpatients with renal impairment.
Hypersensitivity reactions
Levofloxacin can cause serious, potentially fatal hypersensitivity reactions (e.g. angioedema up to anaphylactic shock), occasionally following the initial dose. Patients should discontinue treatment immediately and contact their physician or an emergency physician, who will initiate appropriate emergency measures.
Severe cutaneous adverse reactions
Severe cutaneous adverse reactions (SCARs) including toxic epidermal necrolysis (TEN: also known as Lyell's syndrome), Stevens Johnson syndrome (SJS) and drug reaction with eosinophilia and systemic symptoms (DRESS), which could be life-threatening or fatal, have been reported with Levofloxacin. At the time of prescription, patients should be advised of the signs and symptoms of severe skin reactions, and be closely monitored. If signs and symptoms suggestive of these reactions appear, Levofloxacin should be discontinued immediately and an alternative treatment should be considered. If the patient has developed a serious reaction such as SJS, TEN or DRESS with the use of Levofloxacin, treatment with Levofloxacin must not be restarted in this patient at any time.

Dysglycaemia
As with all quinolones, disturbances in blood glucose, including both hypoglycaemia and hyperglycaemia have been reported, occurring more frequently in the elderly, usually in diabetic patients receiving concomitant treatment with an oral hypoglycaemic agent (e.g., Glibenclamide) or with insulin. Cases of hypoglycaemic coma have been reported. In diabetic patients, careful monitoring of blood glucose is recommended. Levofloxacin treatment should be stopped immediately if a patient reports blood glucose disturbance and alternative non fluoroquinolone antibacterial therapy should be considered.
Prevention of photosensitisation
Photosensitisation has been reported with Levofloxacin. It is recommended that patients should not expose themselves unnecessarily to strong sunlight or to artificial UV rays (e.g. sunray lamp, solarium), during treatment and for 48 hours following treatment discontinuation in order to prevent photosensitisation.
Patients treated with Vitamin K antagonists
Due to possible increase in coagulation tests (PT/INR) and/or bleeding in patients treated with Levofloxacin in combination with a vitamin K antagonist (e.g. Warfarin), coagulation tests should be monitored when these drugs are given concomitantly.
Psychotic reactions
Psychotic reactions have been reported in patients receiving quinolones, including Levofloxacin. In very rare cases these have progressed to suicidal thoughts and self-endangering behaviour, sometimes after only a single dose of Levofloxacin. In the event that the patient develops these reactions, Levofloxacin should be discontinued immediately at the first signs or symptoms of these reactions and patients should be advised to contact their prescriber for advice. Alternative non-fluoroquinolone antibacterial therapy should be considered, and appropriate measures instituted. Caution is recommended if Levofloxacin is to be used in psychotic patients or in patients with history of psychiatric disease used in psychotic patients or in patients with history of psychiatric disease.
QT interval prolongation
Caution should be taken when using fluoroquinolones, including Levofloxacin, in patients with known risk factors for prolongation of the QT interval such as, for example:
- congenital long QT syndrome
- concomitant use of drugs that are known to prolong the QT interval (e.g. Class IA and III antiarrhythmics, tricyclic antidepressants, macrolides, antipsychotics).
- uncorrected electrolyte imbalance (e.g. hypokalemia, hypomagnesemia)
- cardiac disease (e.g. heart failure, myocardial infarction, bradycardia)

Elderly patients and women may be more sensitive to QTc-prolonging medications. Therefore, caution should be taken when using fluoroquinolones, including Levofloxacin, in these populations.
Peripheral neuropathy
Cases of sensory or sensorimotor polyneuropathy resulting in paraesthesia, hypaesthesia, dysesthesia, or weakness have been reported in patients receiving quinolones and fluoroquinolones. Patients under treatment with LEVOFLOXOL® should be advised to inform their doctor and pharmacist prior to continuing treatment if symptoms of neuropathy such as pain, burning, tingling, numbness, or weakness develop in order to prevent the development of potentially irreversible condition.
Hepatobiliary disorders
Cases of hepatic necrosis up to fatal hepatic failure have been reported with Levofloxacin, primarily in patients with severe underlying diseases, e.g. sepsis. Patients should be advised to stop treatment and contact their doctor if signs and symptoms of hepatic disease develop such as anorexia, jaundice, dark urine, pruritus or tender abdomen.
Exacerbation of myasthenia gravis
Fluoroquinolones have neuromuscular blocking activity and may exacerbate muscle weakness in person with myasthenia gravis. Post marketing serious adverse events, including deaths and requirement for ventilator support have been associated with fluoroquinolones use in persons with myasthenia gravis. Avoid fluoroquinolones in patients with known history of myasthenia gravis.
The use of LEVOFLOXOL® should be avoided in patients who have experienced serious adverse reactions in the past when using fluoroquinolones containing products. Treatment of these patients with LEVOFLOXOL® should only be initiated in the absence of alternative treatment options and after careful benefit/risk assessment.
Vision disorders
If vision becomes impaired or any effects on the eyes are experienced, an eye specialist should be consulted immediately.
Superinfection
The use of Levofloxacin, especially if prolonged, may result in overgrowth of non-susceptible organisms. If superinfection occurs during therapy, appropriate measures should be taken.
Interference with laboratory test
In patients treated with Levofloxacin, determination of opiates in urine may give false-positive results. It may be necessary to confirm positive opiate screens by more specific method.
Levofloxacin may inhibit the growth of *Mycobacterium tuberculosis* and, therefore, may give false-negative results in the bacteriological diagnosis of tuberculosis.
Aortic aneurysm and dissection
Epidemiologic studies report an increased risk of aortic aneurysm and dissection after intake of fluoroquinolones, particularly in the older population. Therefore, fluoroquinolones should only be used after careful benefit-risk assessment and after consideration of other therapeutic options in patients with positive family history of aneurysm disease, or in patients diagnosed with pre-existing aortic aneurysm and/or aortic dissection, or in presence of other risk factors or conditions predisposing for aortic aneurysm and dissection (e.g. Marfan syndrome, vascular Ehlers-Danlos syndrome, Takayasu arteritis, giant cell arteritis, Behcet's disease, hypertension, known atherosclerosis). In case of sudden abdominal, chest or back pain, patients should be advised to immediately consult a physician in an emergency department.
LEVOFLOXOL® contains Sodium
This medicinal product contains 354 mg Sodium per 100 mL equivalent to 17.7% of the WHO recommended maximum daily intake of 2 g Sodium for an adult. Levofloxacin solution for infusion is considered high in Sodium. This should be particularly taken into account for those on a low salt diet.

Interactions with Other Medicaments
Effect of other medicinal products on Levofloxacin
Theophylline, fenbufen or similar non-steroidal anti-inflammatory drugs
No pharmacokinetic interactions of Levofloxacin were found with Theophylline in a clinical study. However, a pronounced lowering of the cerebral seizure threshold may occur when quinolones are given concurrently with Theophylline, non-steroidal anti-inflammatory drugs, or other agents which lower the seizure threshold. Levofloxacin concentrations were about 13% higher in the presence of Fenbufen than when administered alone.
Probenecid and Cimetidine
Probenecid and Cimetidine had a statistically significant effect on the elimination of Levofloxacin. The renal clearance of Levofloxacin was reduced by Cimetidine (24%) and Probenecid (34%). This is because both drugs are capable of blocking the renal tubular secretion of Levofloxacin. However, at the tested doses in the study, the statistically significant kinetic differences are unlikely to be of clinical relevance. Caution should be exercised when Levofloxacin is co-administered with drugs that affect the tubular renal secretion such as Probenecid and Cimetidine, especially in renally impaired patients.
Other relevant information
Clinical pharmacology studies have shown that the pharmacokinetics of Levofloxacin were not affected to any clinically relevant extent when Levofloxacin was administered together with the following drugs: Calcium Carbonate, Digoxin, Glibenclamide, and Ranitidine.

Effect of Levofloxacin on other medicinal products
Cyclosporin
The half-life of Cyclosporin was increased by 33% when co-administered with Levofloxacin.
Vitamin K antagonists
Increased coagulation tests (PT/INR) and/or bleeding, which may be severe, have been reported in patients treated with Levofloxacin in combination with a vitamin K antagonist (e.g. Warfarin). Coagulation tests, therefore, should be monitored in patients treated with vitamin K antagonists.
Drugs known to prolong QT interval
Levofloxacin, like other fluoroquinolones, should be used with caution in patients receiving drugs known to prolong the QT interval (e.g. Class IA and III antiarrhythmics, tricyclic antidepressants, macrolides, antipsychotics).
Other relevant information
In a pharmacokinetic interaction study, Levofloxacin did not affect the pharmacokinetics of theophylline (which is a probe substrate for CYP1A2), indicating that Levofloxacin is not a CYP1A2 inhibitor.

Pregnancy and Lactation:
Pregnancy
There are limited amount of data from the use of Levofloxacin in pregnant women. Animal studies do not indicate direct or indirect harmful effects with respect to reproductive toxicity.
However, in the absence of human data and due to that experimental data suggest a risk of damage by fluoroquinolones to the weight-bearing cartilage of the growing organism, Levofloxacin must not be used in pregnant women.
Breastfeeding
Levofloxacin is contraindicated in breastfeeding women. There is insufficient information on the excretion of Levofloxacin in human milk; however, other fluoroquinolones are excreted in breast milk. In the absence of human data and due to that experimental data suggest a risk of damage by fluoroquinolones to the weight-bearing cartilage of the growing organism, Levofloxacin must not be used in breastfeeding women.
Fertility
Levofloxacin caused no impairment of fertility or reproductive performance in rats.

Adverse Effects / Undesirable Effect:
The information given below is based on data from clinical studies in more than 8300 patients and on extensive post marketing experience.
Frequencies in this table are defined using the following convention: very common (≥1/10), common (≥1/100, <1/10), uncommon (≥1/1000, <1/100), rare (≥1/10000, <1/1000), very rare (<1/10000), not known (cannot be estimated from the available data). Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

Table of adverse reactions:

System organ class	Common (≥1/100 to <1/10)	Uncommon (≥1/1,000to<1/100)	Rare (≥1/10,000 to <1/1,000)	Not known (cannot be estimated from available data)
Infections and infestations	-	Fungal infection including Canadida infection, Pathogen resistance	-	-
Blood and the lymphatic system disorders	-	Leukopenia, Eosinophilia	Thrombocytopenia, Neutropenia	Pancytopenia, Agranulocytosis, Haemolytic anaemia
Immune system disorders	-	-	Angioedema, Hypersensitivity	Anaphylactic shock[a], Anaphylactoid shock[a]
Endocrine disorders	-	-	Syndrome of inappropriate secretion of antidiuretic hormone (SIADH)	-
Metabolism and nutrition disorders	-	Anorexia	Hypoglycaemia particularly in diabetic patients, Hypoglycaemic coma	Hyperglycaemia
Psychiatric disorders*	Insomnia	Anxiety, Confusional state, Nervousness	Psychotic reactions (with e.g. hallucination, paranoia), Depression, Agitation, Abnormal dreams, Nightmares, Delirium	Psychotic disorders with self-endangering behaviour including suicidal ideation or suicide attempt
Nervous system disorders*	Headache, Dizziness	Somnolence, Tremor, Dysgeusia	Thrombocytopenia, Neutropenia	Peripheral sensory neuropathy, Peripheral sensory motor neuropathy, Parosmia including anosmia, Dyskinesia, Extrapyramidal disorder, Ageusia, Syncope, Benign intracranial hypertension
Eye disorders*	-	-	Visual disturbances such as blurred vision	Transient vision loss, uveitis
Ear and labyrinth disorders*	-	Vertigo	Tinnitus	Hearing loss, Hearing impaired
Cardiac disorders**	-	-	Tachycardia, Palpitation	Ventricular tachycardia, which may result in cardiac arrest Ventricular arrhythmia and torsade de pointes (reported predominantly in patients with risk factors of QT prolongation), electrocardiogram QT prolonged
Vascular disorders**	Applies to iv form only: Phlebitis	-	Hypotension	-
Respiratory, thoracic and mediastinal disorders	-	Dyspnoea	-	Bronchospasm, Pneumonitis allergic
Gastrointestinal disorders	Diarrhoea, Vomiting, Nausea	Abdominal pain, Dyspepsia, Flatulence, Constipation	-	Diarrhoea – haemorrhagic which in very rare cases may be indicative of enterocolitis, including pseudomembranous colitis, Pancreatitis
Hepatobiliary disorders	Hepatic enzyme increased (ALT/AST, Alkaline Phosphatase, GGT)	-	-	Jaundice and severe liver injury, including fatal cases with acute liver failure, primarily in patients with severe underlying diseases, Hepatitis
Skin and subcutaneous tissue disorders[b]	-	Rash, Pruritus, Urticaria, Hyperhidrosis	Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS), Fixed drug eruption	Toxic epidermal necrolysis, Stevens-Johnson syndrome Erythema, multiforme, Photosensitivity reaction, Leukocytoclastic vasculitis Stomatitis
Musculoskeletal and connective tissue disorders*	-	Arthralgia, Myalgia	Tendon disorders including Tendonitis (e.g. Achilles tendon) Muscular weakness which may be of special importance in patients with myasthenia gravis	Rhabdomyolysis, Tendon rupture (e.g. Achilles tendon), Ligament rupture, Muscle rupture, Arthritis
Renal and urinary disorders	-	Blood creatinine increased	Renal failure acute (e.g. due to interstitial nephritis	-
General disorders and administrative site conditions*	Applies to IV form only: Infusion site reaction (pain, reddening)	Asthenia	Pyrexia	Pain (including pain in back, chest, and extremities)

[a] Anaphylactic and anaphylactoid reactions may sometimes occur even after the first dose
[b] Mucocutaneous reactions may sometimes occur even after the first dose

Other undesirable effects which have been associated with fluoroquinolone administration include:
- attacks of porphyria in patients with porphyria.
* Very rare cases of prolonged (up to months or years), disabling and potentially irreversible serious drug reactions affecting several, sometimes multiple, system organ classes and senses (including reactions such as tendinitis, tendon rupture, arthralgia, pain in extremities, gait disturbance, neuropathies associated with paraesthesia, depression, fatigue, memory impairment, sleep disorders, and impairment of hearing, vision, taste and smell) have been reported in association with the use of fluoroquinolones in some cases irrespective of pre-existing risk factors.
** Cases of aortic aneurysm and dissection, sometimes complicated by rupture (including fatal ones), and of regurgitation/incompetence of any of the heart valves have been reported in patients receiving fluoroquinolones.

Exacerbation of myasthenia gravis
Post Marketing Experience

Overdose and Treatment:
According to toxicity studies in animals or clinical pharmacology studies performed with supra-therapeutic doses, the most important signs to be expected following acute overdose of LEVOFLOXOL® are central nervous system symptoms such as confusion, dizziness, impairment of consciousness, and convulsive seizures, increases in QT interval. CNS effects including confusional state, convulsion, hallucination, and tremor have been observed in post marketing experience. In the event of overdose, symptomatic treatment should be implemented. ECG monitoring should be undertaken, because of the possibility of QT interval prolongation. Haemodialysis, including peritoneal dialysis and CAPD, are not effective in removing Levofloxacin from the body. No specific antidote exists.

Effects on ability to drive and use machines
Levofloxacin has minor or moderate influence on the ability to drive and use machines. Some undesirable effects (e.g. dizziness/vertigo, drowsiness and visual disturbances) may impair the patient's ability to concentrate and react, and therefore may constitute a risk in situations where these abilities are of special importance (e.g. driving a car or operating machinery).

Incompatibilities:
LEVOFLOXOL® should not mixed with heparin or alkaline solutions (eg: Sodium Hydrogen Carbonate).

Caution:
Do not use if bottle is leaking, solution is cloudy or contains particle. Single use only. Discard any unused portion.

Storage Conditions:
Do not store above 30°C. Protect from light.

Shelf Life:
2 years from manufacturing date in the proposed storage condition. Do not use after expiry.

Dosage Forms and Packaging Available:
100 mL X 20 LDPE plastic bottle per carton.

Manufacturer / Product Registration Holder:

AIN MEDICARE SDN. BHD.
Lot 4933, 4934 & PT2464, Jalan 6/44,
Kaw. Perindustrian Pengkalan Chepa 2,
16100 Kota Bharu, Kelantan, MALAYSIA

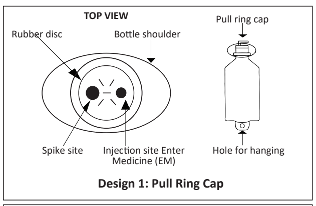
Date of Revision : 24.01.2024

HANDLING INSTRUCTION

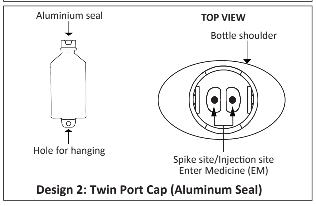
AIN MEDICARE INTRAVENOUS INFUSION FLUID IN LOW DENSITY POLYETHYLENE (LDPE) BOTTLE
The bottle is made from injectable grade polyethylene.

PARTS OF THE PLASTIC BOTTLE

Top view after pull ring cap is removed. Please refer whichever applicable design.

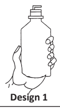


Design 1: Pull Ring Cap

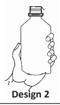


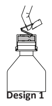
Design 2: Twin Port Cap (Aluminum Seal)

INSTRUCTION FOR THE USAGE OF IV ADMINISTRATION SET & ADDITION OF DRUG

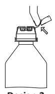

Design 1

1. Before use check for any solid particle, growth, turbidity or leakage. Discard the product if particle, growth, turbidity or leakage is found.


Design 2


Design 1

2. Design 1: Pull the cap's ring until completely removed. Design 2: Pull the aluminum seal until completely removed by centre line.


Design 2


Design 1

3. Additives may be injected through injection site (Refer to Design 1 or 2 whichever applicable) on the rubber. Mix the solution thoroughly.


Design 2


Design 1

4. Hold bottle down firmly and insert a spike of administration set into the spike site for Design 1 OR for Design 2, pull the other aluminium seal until completely removed before inserting a spike of administration set. *(Spike site is meant for spiking of the administration set and is only for single use)*


Design 2